

| 10" PE BALL VALVE COMPARISONS                   |       |                          |                                |               |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|-------|--------------------------|--------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | Pipe  | Polyvalve for Water      | Georg Fischer/Central Plastics | Lyall/Polytec | Polyvalve is the Nordstrom PE ball valve. Lyall is the agent for the Korean firm Polytec and supplies CMF and Integrity Fusion this valve as a private label. The Central Plastics valve is made by a Chinese manufacturer selling to Georg Fischer.                                                                                                                                                                                   |
| Country of Origin                               |       | USA                      | China                          | Korea/China   | Polytec/Lyall has manufacturing operations in China and Korea.                                                                                                                                                                                                                                                                                                                                                                         |
| Year of first USA use                           |       | 1976                     | ~2010                          | ~ 2004        | Polyvalve/Nordstrom was the inventor of this type of valve. Exact date of Central Plastics valve is not clear. They were looking at these when I visited them in 2008 but had not yet commercialized them. Lyall introduced this valve in about 2004. Their literature shows 1993 which I presume was when the Korean firm first started making this valve or its predecessor.                                                         |
| Primary use                                     |       | Water                    | Gas                            | Gas           | The experience of the Polyvalve offering is entirely water applications. The experience of the Lyall valve is almost entirely gas. The Central valve is primarily used in gas applications but it is very new. The gas market is the biggest market for PE valves. Keep in mind that gas systems operate at lower pressures than golf systems and these same valves carry pressure ratings about half of those used for water systems. |
| ID of flow way                                  | 8.796 | 8.889                    | 8.681                          | 7.95          | The Central and Lyall valves are adaptations of metric valves. The Lyall valve chokes the flow by approximately 1" in flow diameter. The Central valve is about the same diameter as pipe.                                                                                                                                                                                                                                             |
| Stem material                                   |       | Modified Phenylene Oxide | ABS                            | Acetal        | The Polyvalve material is chosen in lieu of the Acetal typically used in gas valves because of Acetal's vulnerability to small concentrations of Chlorine compounds used in water disinfection. The ABS used by Central is very unconventional for which there is no precedent in this country.                                                                                                                                        |
| Ball & Retainer material                        |       | Polypropylene            | Nylon                          | Polypropylene | Polyvalve and Lyall's use of Polypropylene are typical and of long experience. Central's use of Nylon is unusual and for water is questionable since Nylons will swell in the presence of water which could effect actuation in the long term.                                                                                                                                                                                         |
| Number of Stem Seals                            |       | 3                        | 2                              | 2             |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Stem Seal material                              |       | EPDM                     | Nitrile                        | Nitrile       | EPDM or SBR are ideal for water but not gas. Nitrile is ideal for gas and when optimized for gas is suboptimized for water including the ability to pass toxicity tests.                                                                                                                                                                                                                                                               |
| Body material                                   |       | 4710 resin               | 4710 resin                     | 3408 resin    | Lyall does not use the most modern resins and as a result is not as highly pressure rated. Polyvalve uses the most advanced Dow bimodal resin. Central uses a bimodal resin but the exact one is not identified.                                                                                                                                                                                                                       |
| Pressure rating of body                         |       | 250 psi                  | 200 psi                        | 160 psi       | This is a function of wall thickness and resin.                                                                                                                                                                                                                                                                                                                                                                                        |
| Standard Actuation                              |       | Gear                     | Gear                           | Gear          | Polyvalve and Central can only be had with gear actuators. Lyall is standard with an actuator but can be had with a simple square nut with a lead time and delivery from Asia.                                                                                                                                                                                                                                                         |
| Gear actuator materials                         |       | 316 SS                   | Plastic                        | Plastic       | The Lyall valve gears are plastic with brass trim. The type of brass is unclear but possibly yellow brass. The Central valve is plastic with some sort of metal trim. The customer service rep at Central did not know what type of metal the trim was. These plastic gear actuation systems are not accepted by all gas distribution systems.                                                                                         |
| Rise from main to top of actuator               |       | ~11"                     | ~17.5"                         | ~18"          |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Ball lubricant                                  |       | For Water                | For Gas                        | For Gas       | The Polyvalve lubricant is optimized for water. Lubricants optimized for gas have lower effective lifetimes in the presence of water typically. Also when optimized for gas they can be challenging or impossible to pass toxicity requirements for water systems.                                                                                                                                                                     |
| Potable Water Toxicity Approval, NSF 61 Listing |       | Yes                      | Yes                            | No            | NSF 61 listing does not address suitability for use in the water application for other than toxicological risk. It does not address any other performance consideration. Therefore one should not infer a product is suitable for a water application in all respects solely because of this listing.                                                                                                                                  |